PRIYANKA GONUGUNTA

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PROFESSIONAL SUMMARY

Senior Data Scientist and Machine Learning Engineer with 10+ years of experience delivering predictive analytics, recommendation systems, financial analytics modernization, scalable data engineering, and cloud-based ML solutions. Strong Python, SQL, AWS SageMaker, Snowflake/Snowpark and MLOps background, including model training/deployment, regression validation, CI/CD, and production SAS/Netezza migrations. Applied Generative AI assistants and LLM-based coding tools to accelerate engineering, SQL conversion, testing, and documentation at Toyota, validating outputs through review and regression checks. Familiar with GenAI/RAG architecture, embeddings, prompt engineering, retrieval grounding, and agentic AI concepts; no production agent-platform implementation claimed.

TECHNICAL SKILLS

Programming & Data Python, SQL, SAS, R, Pandas, PySpark, Apache Spark, Snowflake, Snowpark, Netezza, NoSQL, ETL/ELT, data modeling

ML / Analytics Regression, classification, clustering, RFM, recommendations, feature engineering, time series, A/B testing, hypothesis testing, SMOTE, AUROC, TensorFlow, PyTorch, NLP, computer vision, BERT, RoBERTa

Generative AI & Agentic AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Sentence Transformers, Vector Embeddings, Semantic Search, Cosine Similarity, Ollama, Llama 3.2, LangGraph (Prototype), Tool Calling, Agent Orchestration Fundamentals, MCP Concepts, LLM Evaluation.

Cloud / MLOps / BI : AWS SageMaker, S3, IAM, EC2, Redshift, MLflow, Docker, FastAPI, Git/GitHub, CI/CD, YAML, pytest, Power BI, Tableau, MicroStrategy, JIRA, Confluence, VS Code

PROFESSIONAL EXPERIENCE

Toyota Financial Services | Data Scientist / Analyst | *Nov 2025 – Oct 2026*

- Modernized SAS/Netezza financial analytics into modular Python and Snowflake/EDP production pipelines, improving maintainability, automation, source-to-target consistency, and scalable processing.
- Enhanced the Post-Booking Cashflow Engine for retail and lease portfolios, integrating preprocessing, loss curves, account-level data, financial calculations, and downstream ALM/Treasury reporting.
- Re-engineered loss-curve pipelines using vectorized Snowflake SQL and Python/Pandas, reducing intermediate datasets, pivoting, memory overhead, and data-loading bottlenecks.
- Migrated and validated ALM/Treasury curve-generation and forecast workflows; investigated variances by LTV, FICO, months on book, vintage, segmentation, and curve mappings.
- Built pytest unit/regression checks, parity reconciliations, and validation routines for complex financial rules; contributed to SageMaker-based analytics, YAML configuration, Git branching, and CI/CD practices.
- Leveraged Generative AI tools (ChatGPT and Claude) to accelerate legacy SAS-to-Python modernization, SQL query conversion, code refactoring, debugging, automated unit-test generation, and technical documentation; ensured reliability through code reviews, pytest regression testing, and validation against legacy system outputs.

SKILLS:
SAS, Python, Snowpark, Snowflake, JIRA, Confluence, Git, Sage maker, Generative AI, Large Language Models (LLMs), Prompt Engineering, LLM Integration, Automated Code Generation, Code Modernization, pytest, Unit Testing, Regression Testing, Git, MLOPS

Toyota Financial Services | Machine Learning Engineer | *Apr 2025 – Oct 2025*

- Migrated more than 30,000 lines of legacy SAS code into reusable Python/Snowflake pipelines; reported 40% lower code complexity and more than 70% less manual intervention through modular automation.
- Integrated AWS SageMaker training and deployment workflows to support ML lifecycle automation, with a reported 60% improvement in retraining efficiency.
- Developed automated regression and accuracy tests achieving approximately 99% parity with legacy outputs, cutting QA turnaround from days to hours.
- Implemented YAML-driven configuration, version control and CI/CD-aligned release practices; improved large-volume transformations through optimized Snowpark and SQL processing (reported 30–40% gains).

SKILLS:
SAS, Python, VScode, Snowpark, Snowflake, JIRA, Confluence, Git, Sage maker.

Landmark Group | Senior Data Scientist | *Nov 2018 – Dec 2022*

- Led a six-person analyst team supporting Sales, Marketing and Category functions with customer intelligence, promotional measurement, campaign optimization and actionable retail performance analysis.
- Developed K-means and RFM segmentation, response propensity and predictive marketing models to personalize campaigns; reported AED 150M incremental revenue and 30% improved campaign effectiveness.
- Designed image-processing and visual similarity recommendation algorithms for Splash Apparel, including color-based product matching that increased product engagement by 25% and received internal recognition.
- Applied exploratory analysis, experimentation, A/B testing, and statistical evaluation to inform category-level decisions, improve targeting, and increase campaign ROI by a reported 20%.

- Built Python/TensorFlow and AWS SageMaker predictive modeling workflows; used Spark-based data processing to transform large retail datasets and supply model-ready features.
- Presented model findings to cross-functional business leaders and converted insights into recommendations for product assortment, sales opportunities, and customer engagement.

SKILLS:

Predictive Modeling, Clustering, CNN, CV, R, Python, SAS, SQL, Microsoft Excel (Advanced), Tableau, AWS.

Sybrant Technologies | Senior Data Analyst | *Aug 2015 – Dec 2017*

- Analyzed large financial and customer datasets using SQL, Python and SAS; reconciled inputs, identified anomalies, and prepared reliable data for reporting and predictive analysis.
- Automated daily, weekly and monthly extracts and recurring reports with SAS macros, reducing repetitive preparation and improving reporting consistency.
- Developed loan-default and customer-churn classification models using logistic regression and decision trees, supporting risk segmentation and faster operational decisions.
- Applied K-means clustering, SMOTE for class imbalance, hypothesis testing, and model evaluation to strengthen risk profiling and assess predictive performance.
- Created Power BI, Tableau and MicroStrategy dashboards and ad hoc KPI reporting for management, finance and operational stakeholders.

SKILLS:

SAS, SQL, Machine Learning, MicroStrategy, Campaigns, Adhocs, Microsoft Excel (Advanced), PowerBI, Tableau.

Botcode Integration | Data Analyst | *Aug 2012 – Jul 2015*

- Collected and integrated structured data from multiple business sources using SQL and Python; cleaned records, standardized fields, and checked data accuracy before analysis.
- Built automated extraction, transformation, and recurring reporting routines that improved data availability and reduced manual preparation for business teams.
- Performed exploratory data analysis, regression, basic clustering and time-series analysis to identify customer trends, operating patterns, and notable changes in performance.
- Created interactive Power BI/Tableau dashboards to track business KPIs, compare historical results, and communicate findings to nontechnical stakeholders.
- Produced customer-segmentation and campaign reporting to support targeting decisions, marketing reviews, and ROI analysis; documented reporting logic and data definitions.

SKILLS:

SAS, SQL, Machine Learning, MicroStrategy, Campaigns, Adhocs, Microsoft Excel (Advanced), and Microsoft PowerPoint, Tableau, Power BI.

INDEPENDENT GENERATIVE AI & AGENTIC AI PROJECTS

Enterprise Financial Intelligence Copilot | Generative AI, RAG & Agentic AI

- Developed an AI-powered financial intelligence assistant to simplify access to financial risk, loan prepayment, and portfolio analytics documentation, enabling natural-language querying through semantic retrieval and LLM-generated responses.
- Engineered an end-to-end semantic search pipeline using document chunking, 384-dimensional Sentence Transformer embeddings, cosine similarity, and top-k vector retrieval to identify relevant financial information beyond conventional keyword searches.
- Integrated Retrieval-Augmented Generation (RAG) with Ollama and Llama 3.2, incorporating prompt engineering and retrieved document context to generate grounded responses with source references and improve answer traceability.
- Built a LangGraph-based agent prototype integrating semantic document retrieval and financial calculation tools, with a design supporting LLM-driven tool calling, conversation state management, and multi-step agent orchestration.
- Evaluated RAG response reliability through functional testing, similarity-score analysis, source verification, and missing-information scenarios, demonstrating accurate retrieval and appropriate handling of unsupported financial queries.

SKILLS:

Python, Sentence Transformers, Vector Embeddings, RAG, Ollama, Llama 3.2, Streamlit, Lang Graph (Prototype)

EDUCATION

- MS in Data Science, Webster University – CGPA: 3.5/4 (Dec'2024)
- MS in Business Analytics, Praxis Business School – CGPA: 3.0/4 (2019)
- MS in Information Technology, Anna University – CGPA: 3.8/4 (2012)

AWARDS & RECOGNITION

- Appreciation for Developing a Color-based Recommendation algorithm for the Splash Apparel Data.
- Saviour Award – for additional responsibilities when the team was short on resources.
- Exceptional Support to CI GCC award in the Landmark group.
- Spot Award for expeditiously developing the Ramadan Dashboard in Landmark Group.
- Ranked 7th in the Information Technology department at Anna University with an overall CGPA of 3.8.
- Participated in Hackathon conducted by Machine Hack (2018), Leader board position of Top 20th Percentile.